

Surprise without the field: what the question means, what can answer it, and a card to run

2026-09-02, tier-1 session, for query R2.Q5 as Jonas narrowed it on 2026-08-29: before the active-inference contribution is positioned in any paper, explore whether the surprise elements can be kept on their own, with surprise defined as a scenario-agnostic metric, even if the preference field is dropped. This is a design note, not an analysis: nothing new is computed here, and every number quoted comes from a tracked output named at the point of quotation. It ends with one executable card (Q5.1) for the cheaper tier, with its decision rules stated in advance, and with what I think each outcome would mean for positioning. Opinions are marked as such.

0 The answer in one paragraph

The way I read it, "surprise without the field" is a well-posed question only once one says what the surprise is relative to, because every member of the surprise family is an operator on a reference distribution and the operator itself carries almost no content: for any non-negative score there is a reference under which that score *is* residual information. Scenario-agnosticism is therefore a property of the reference, not of the operator, and there are exactly two classes of reference that are scenario-agnostic by construction. One is a **predictive model of what other road users normally do**, which makes surprise a statement about the world (the Waymo construction). The other is a **population distribution of the states drivers normally accept**, which makes surprise a statement about the ego's situation relative to its peers, and which turns out to be the CZB ellipse under another name: the Mahalanobis joint percentile is residual information under a Gaussian population reference. The first cannot be the comfort-zone criticality axis on its own, because it is zero by construction in scenarios where the other agent behaves normally and the discomfort comes from the ego's own trajectory (the cyclist overtake, one of the four scenarios on which the per-driver trait is shared). It may still do a job the field was supposed to do and the gap threshold cannot: define *when the stimulus begins*, independent of the lateral pace of the manoeuvre, which is one of the two constraints the pipeline review extracted from the data. So the candidate answer to R2.Q5 is not "surprise as the axis" but "surprise as the onset, population percentile as the level": two references, not one field read at two levels. Card Q5.1 tests the onset half on data already in hand, with no preference function anywhere in the computation.

1 What the project's surprise was, and what exactly was falsified

The project used one member of one family. Residual information,

$$h(x; P) = \log \max_{x'} P(x') - \log P(x) \geq 0,$$

applied to the **preference prior** $p(o)$ of the published model (Schumann et al., 2026) gives the pragmatic deficit: surprise about one's own preferred state. Evaluated pointwise along a recorded trajectory it was the comfort-zone field; accumulated over time it was the response-timing mechanism. Both readings have been tested on human data and both failed their pre-registered rules (docs/r2_gate_decisions.md; the accumulator at gate R.1, replication/czb/out/stage2_summary.md; the axis at gate R.2, replication/czb/out/cutin2_field_vs_gap.md).

The pipeline review (docs/r2_pipeline_review.md) says what failed more precisely, and it matters here. The field on the decisive cells was, to a good approximation, a lane-entry gate times a worst-case counterfactual magnitude. The gate, a project construction, made the deficit depend on the lateral pace of the cut-in when participants do not (their response is flat across 2, 3 and 4 s lane changes at matched time since onset, TTC and delta velocity); the magnitude, the published model's "lead brakes at 6 m/s², I react after 1 s" counterfactual, ordered matched-TTC cells by the vehicles' absolute speeds when participants order them by gap. Neither failure is a failure of the residual-information operator, which did what it does: it reported how far the state was from the reference's mode. Both are failures of the **reference**.

What was *not* used is worth listing, because it is what R2.Q5 asks about. The library src/surprise/ implements three further families (Modirshanechi et al., 2022; Dinparastdjadid et al., 2023): probabilistic mismatch of an observation against a predictive belief (surprisal, residual information, Bayes-factor surprise), belief mismatch between a prior and a posterior belief (Bayesian surprise, postdictive surprise, antithesis, confidence-corrected surprise), and observation mismatch against a point prediction. All of them take a *predictive* distribution about the world as their reference. None of them has been evaluated against any behavioral data in this project. The library is validated on its own terms (31 property tests) and works unchanged on Gaussians, mixtures, particle sets and

categoricals; that is a statement about software, not about drivers.

2 Why the reference is the whole question

A short argument that I think settles what "keep surprise, drop the field" can and cannot mean.

Take any non-negative score $s(x)$ on states, for instance "gap threshold exceedance", $s = \max(c - \text{gap}, 0)$. Define a reference $P(x) \propto \exp(-s(x))$. Then residual information under P is $h(x; P) = s(x) - \min s = s(x)$. So the surprise family with a *free* reference reproduces every threshold model there is, including the log-gap threshold that beat the field. It follows that a proposal of the form "replace the preference prior by a different hand-built prior over the ego's states" is a relabeling of a threshold model, and gains nothing from the surprise vocabulary except the zero floor, which the threshold model already has. I would not put such a proposal in a paper under the surprise name, and I do not think Jonas's question is asking for it.

The content of a surprise measure comes entirely from constraints on the reference that one is not free to choose: that it is a genuine *predictive* distribution, estimated from how road users actually behave, or a genuine *population* distribution, estimated from what drivers actually accept. Under either constraint the reference is fixed by data rather than by the analyst, the operator then has something to say, and, decisively for R2.Q5, both constraints are **scenario-agnostic in form**. A predictor of other agents' motion does not know what scenario it is in; a population distribution of accepted states can be estimated for any scenario by the same recipe. The preference prior, by contrast, was scenario-agnostic in its six terms but scenario-specific in what the two conflict terms encode, which is where it lost.

The rest of this note takes the two constrained references in turn.

3 Reference A: a predictive model of the world

3.1 What it is

This is the construction of Dinparastdjadid et al. (2023): a generative model emits, at each time t , a belief about each other agent's state at future times; surprise is the observation's residual information under the belief made h seconds earlier (probabilistic mismatch), or the divergence between the belief made at $t - h$ and the belief made at t about the same future time $t + z$ (belief mismatch; the paper's own measure, antithesis, is the one that stays zero when nothing unexpected happens). The library implements exactly this bookkeeping (`surprise.timeseries.surprise_time_series`), including the two timing parameters, which the paper reports must be swept because the effect of z is not monotone.

Two facts about this reference decide what it can do for us.

It is scenario-agnostic by construction. The same predictor, run on the other agent's recorded positions in a body frame, yields a lateral and a longitudinal surprise series in a cut-in, a left turn across path, a truck drifting laterally, or a cyclist being passed. Nothing in it needs the hand geometry that every field construction has needed (lane gates, crossing zones, co-occupancy rules).

It is zero wherever the other agent does nothing unexpected. That is not a defect; it is what "surprise about the world" means. But it means that in the cyclist overtake, where the cyclist rides steadily and the ego chooses how close to pass, world surprise is identically zero at every clearance, while participants intervene at 0.14 to 0.69 across the 15 cells (`replication/czb/out/transfer_overtake_summary.md`). The same holds for any scenario in which discomfort arises from the ego's own trajectory. Since the per-driver comfort-zone level is shared at about 0.69 of the reliability ceiling across all four scenarios *including* the overtake (`replication/czb/out/cross_scenario_consistency.md`), whatever the shared trait is, it is not sensitivity to world surprise alone. This is, as far as I can tell, a derivation rather than a finding, and it rules out reference A as the sole criticality axis before anything is run.

3.2 What it may still do: define the onset

The pipeline review extracted two constraints from the second cut-in study on any reference that would replace the field. The first was that the response engages when encroachment *begins*, independent of how fast the lateral motion proceeds. A belief-mismatch surprise about the target's lateral position has precisely that property: its onset is the first frame at which the target's lateral motion departs from the prediction, which does not depend on the pace of the motion, while its *magnitude* does (a faster lane change is a larger departure). The field's gate had the opposite

property: its onset depended on the pace, through the projection to closure.

So the honest role of reference A, in my reading, is the one the Waymo paper itself proposes for it: to define when the stimulus begins (their second application, response-time modeling, "solving the when-do-you-start-the-clock problem"), and, in combination with a proximity condition, to define when a conflict exists (their first application: conflict = surprise $\square$ spatiotemporal proximity). Neither of those is the criticality axis. The axis, on the decisive data, is proximity, currently best represented by the log gap and next by TTC; the surprise says when the axis starts to count. That is a coherent two-object account, and it is consistent with the trait finding: the shared per-driver level is a proximity tolerance, and the scenario-specific third of the variance may in part be how surprising the different scenarios' other agents are.

3.3 What predictor, given what we have

We have no learned predictor of normal driving, and the stimuli are simulator-scripted, so the reference cannot be fitted to the study clips themselves without circularity. The stand-in that costs nothing and is standard in the trajectory-prediction literature as the baseline is a **constant-velocity Gaussian predictor**: the other agent's future body-frame position is Gaussian around its constant-velocity extrapolation, with standard deviation growing with the horizon. This has two parameters per axis (the uncertainty at zero horizon and its growth rate), plus the paper's h and z . Every one of them needs a stated motivation before the card runs; the card below states them as sweeps with the paper's advice as the motivation, and requires the report to show the sensitivity rather than one chosen value. A constant-velocity reference is also honest about what it claims: it encodes only "road users keep doing what they are doing", which is the weakest possible norm and is the same in every scenario.

Two better references exist and are named for the record. A predictor learned from naturalistic data, which is what the NDS access is partly for (docs/czb_validation_roadmap.md §4c); and the published model's own norm-shaped prediction machinery, run open-loop on recorded kinematics (the particle filter and the norm weighting without the planner), which would make the reference the model's beliefs rather than a kinematic prior. The second inherits the per-scenario hand-written norms of handbook chapter 07, and so is scenario-agnostic in machinery but not in content; it is the natural follow-up if the constant-velocity stand-in shows the onset property and nothing more.

4 Reference B: a population distribution of accepted states, which is the ellipse

The second scenario-agnostic reference is the distribution of states drivers routinely accept: steady-state headways at speed, accepted lateral clearances, accepted arrival-time separations. Surprise relative to that reference asks "how unusual is my present state among the states drivers normally tolerate", which is a sensible reading of what a comfort-zone boundary is, and the roadmap already lists its estimation as one of the uses of naturalistic data (§4c, use 3: "drivers should mostly live inside their own comfort zones").

A small identity shows that this reference is not a new proposal but the one Jonas has already promoted. If the population of accepted states in a scenario's chosen observables is modeled as Gaussian with mean μ and covariance Σ , residual information is

$$h(x) = \frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) = \frac{1}{2} d^2(x),$$

half the squared Mahalanobis distance, and the joint percentile of the CZB ellipse is the χ^2 probability of $d^2(x)$, a monotone function of h . So the **CZB ellipse is the surprise family with a population reference**: the operator (residual information) and the per-driver level are shared across scenarios, and only the reference (μ , Σ , and the choice of observables) is per-scenario. That is exactly the division of labor the scope map's crux asked for, and it gives the ellipse a reading that the multivariate "exceed both percentiles" rule never had: the level set is the set of states equally unusual relative to what drivers accept.

Two consequences for the ellipse design note (queued as the next design task): the per-scenario observables should be the ones whose population distribution can actually be estimated (from the study cells now, from NDS later), and the Gaussian assumption is a choice that the library does not force, since residual information under a mixture or a particle set is available unchanged if a scenario's accepted-state distribution is not elliptical.

5 What this implies for positioning, stated as options

I would not settle the positioning before card Q5.1 has run, but the options can be written down now, and the first sentence of each is true whatever the card shows.

- 1 **Active inference supplied a candidate axis, and the axis was falsified.** True on the record. The paper is then a comparative study of criticality axes with a psychometric estimator, and the surprise family appears only in the history.
- 2 **Surprise about the world defines the stimulus onset; a population percentile defines the level.** True as a construction; whether it is *supported* is what Q5.1 tests (onset independence from lateral pace on study 2; zero on the overtake by construction). If supported, the surprise family keeps an honest, testable role and the paper can say so without claiming the axis.
- 3 **Surprise relative to the population is the boundary.** True as an identity (§4); its empirical standing is the ellipse program's, not a separate claim.

What none of the options can say, in my opinion, is that one scalar does two jobs. That claim is gone with the field, and the two-reference account is the replacement: onset and level are different objects with different references, and they happen to share an operator.

6 Card Q5.1: world surprise on the data in hand, with no preference function

For the cheaper tier, after review of this note. Standing rules of docs/czb_work_orders.md §2 apply. Nothing in src/aidriver/preferences.py is touched or called.

Build. replication/czb/world_surprise.py, importing only surprise.* and the trace loaders. For each stimulus trace (study 1's four scenarios; study 2's 108 cut-in traces; the 15 CAMP rear-end clips as a check on lead braking), compute at every frame t the other agent's body-frame position and a constant-velocity Gaussian belief made at $t - h$ about t (probabilistic mismatch, residual information) and about $t + z$ (belief mismatch, antithesis), with $\sigma(k) = \sigma_0 + \sigma_1 k \Delta t$ on each axis. Report lateral and longitudinal series separately, as the paper does.

Parameters, each with its motivation. $h \in \{0.5, 1.0, 2.0\}$ s and $z \in \{0, 1, 2\}$ s, swept because the paper reports the h effect as monotone and the z effect as not, and gives no scenario-independent recommendation; $\sigma_0 = 0.1$ m on both axes (the traces' positional resolution, from the simulator's 10 Hz output); $\sigma_1 \in \{0.5, 1.0, 2.0\}$ m/s, the range over which a constant-velocity extrapolation's error grows in the trajectory-prediction literature (marked *unverified*: I have not checked a source for this range in this session, and the card should either cite one or treat the sweep as the motivation). No parameter is fitted to any response.

Test 1, onset (the constraint from the review). For every study-2 trace, the first frame at which the lateral belief-mismatch surprise exceeds zero, against the manoeuvre-onset frame from the filename stamps. Pre-stated expectation: the onset lag is within one frame and does not depend on lane-change duration (2, 3, 4 s) at any starting TTC or delta velocity. Report the lag table by (LCD, TTC, DV). A dependence on LCD of more than one frame falsifies the onset claim of §3.2 for this reference.

Test 2, the overtake is zero by construction. On the three cyclist-overtake traces, the world-surprise series on the cyclist. Pre-stated expectation: identically zero (up to σ_0) at every clearance. This is a check that the derivation of §3.1 holds in the code, not a hypothesis; a nonzero series would mean the cyclist's motion is not what the context files describe and the derivation must be re-examined.

Test 3, does surprise carry any axis information beyond gap. On the 288 decisive cells of study 2, a new comparison script (not the registered one, which is not to be modified) with the registered fit, folds and metric, adding two covariates: the running maximum of the longitudinal residual-information surprise over the shown window, and its product with the inverse gap at the window end (the paper's "surprise and proximity" conflict condition made continuous). Decision rule, stated now: if the surprise-alone covariate's held-out wRMSE is more than 0.01 above the gap's 0.152, surprise is not the axis on this data (expected, per §3.1); if the product covariate is within 0.01 of the gap or below it, the two-object account of §3.2 is supported at the level of the axis and should be carried into the ellipse design; if the product is more than 0.01 above the gap, the onset role survives only if Test 1 passed, and surprise contributes nothing to the level.

Test 4, the CAMP check. On the eight braking-lead clips, the longitudinal surprise onset against the scripted brake onset at simulation time 10 s; on the seven constant-speed clips, the expectation is a flat series. This is the rear-end analogue of Test 1 and the one scenario where the published model's own home ground and a human press time coexist in our data.

Output. out/world_surprise.md with the four tests, the parameter sweep as tables, one figure per scenario (the series with the onset marked), and the card's verdicts against the rules above. Suite before and after. Property tests for the predictor (zero surprise on a constant-velocity trace; onset at the first frame of a step in lateral velocity;

invariance to the body-frame choice) in `tests/test_surprise.py`'s style.

What is deliberately not in the card. Any per-scenario geometry, any preference term, any fit of the predictor to responses, and any claim about LTAP or the truck beyond computing their series for inspection: the LTAP trace loader is B.3.v2's job and the truck's B.2's, and this card should not front-run their construction notes. If the generic loader written for this card is good enough for those scenarios, that is a finding to hand to those cards, not a license to score them here.

7 Where I may be wrong

- The argument of §2 (a free reference makes the operator vacuous) is exact, but it could be read as too dismissive of the *practical* content the library adds: the zero floor, the parameter-free treatment of densities, and correct handling of mixtures and particle sets. Those are real, and they are why the library is worth keeping regardless of how R2.Q5 is settled. They are not, in my view, scientific claims about drivers.
- §3.1's derivation that world surprise is zero on the overtake assumes the cyclist's motion is steady in the traces. Test 2 checks this in code before anything is read into it.
- The constant-velocity stand-in is the weakest possible reference, and a negative result on Test 3 with it would not rule out a learned predictor; it would only say that "keeps doing what it is doing" carries no axis information beyond gap on this design. A positive result would be the more informative outcome, and I expect the negative one.
- The identity in §4 is for a Gaussian population; if a scenario's accepted-state distribution is strongly non-elliptical the ellipse and the surprise reading part ways, and the surprise reading is the more general of the two.
- The onset property of belief-mismatch surprise (§3.2) depends on h and on the predictor's uncertainty growth; with a very wide σ_1 the onset is delayed, which is why Test 1 sweeps rather than chooses.

References

Dinparastdjadid, A., Supeene, I., & Engström, J. (2023). *Measuring surprise in the wild* (arXiv:2305.07733). arXiv. <https://arxiv.org/abs/2305.07733> — read from the source PDF by an earlier session; summary in `notes/01_paper_summaries.md` §2.

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